UNIT WORTH OF RESOURCE

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Primal and Dual

Primal:

Maximize $z=\sum_{j=1}^{n} c_j x_j$

$$s.t.\ \sum_{j=1}^{n} a_{ij} x_j \le b_i,\ i=1,2,...,m$$

$$x_j \ge 0, j=1,2,...,n.$$

Standard primal:

Maximize $z=\sum_{j=1}^{n} c_j x_j$

$$s.t.\ \sum_{j=1}^{n} a_{ij} x_j + S_i = b_i,\ i=1,2,...,m$$

$$x_j \ge 0, j=1,2,...,n; S_i \ge 0, i=1,2,...,m$$

Dual:

Minimize $w=\sum_{i=1}^{m} b_i y_i$

$$s.t.\ \sum_{i=1}^{m} a_{ij} y_i \ge c_j,\ \ j=1,2,...,n$$

$$y_i \ge 0, i=1,2,...,m.$$

An example

Primal

Maximize $z = 5x_1 + 7x_2 + 4x_3$

$$s.t.\ \ x_1 + 2x_2 + x_3 \le 5,$$
$$2x_1 - x_2 + 3x_3 \le 4,$$
$$x_1, x_2, x_3 \ge 0.$$

Here, $c_1 = 5, c_2 = 7, c_3 = 4, b_1 = 5, b_2 = 4.$

$a_{11} = 1,\ a_{12} = 2,\ a_{13} = 1$

$a_{21} = 2,\ a_{22} = -1,\ a_{23} = 3.$

$m = 2, n = 3.$

Dual

Minimize $w = b_1 y_1 + b_2 y_2 = 5y_1 + 4y_2$

$$s.t.\ \sum_{i=1}^{2} a_{ij} y_i = a_{1j} y_1 + a_{2j} y_2 \ge c_j,\ j = 1, 2, \ldots, n$$
$$y_1, y_2 \ge 0.$$

Expanding constraints:

$a_{11} y_1 + a_{21} y_2 \ge c_1 \Rightarrow y_1 + 2y_2 \ge 5,$

$a_{12} y_1 + a_{22} y_2 \ge c_2 \Rightarrow 2y_1 - y_2 \ge 7,$

$a_{13} y_1 + a_{23} y_2 \ge c_3 \Rightarrow y_1 + 3y_2 \ge 12.$

Primal

- Variables n
- Resources m
- Profit per unit variable j is c_j
- Consumption of resource i per unit of variable j is a_{ij}

From Property II

$$z=\sum_{j=1}^{n} c_j x_j \le \sum_{i=1}^{m} b_i y_i = w$$

$$\text{Money}=\sum_{j=1}^{n}(\text{profit per unit of variable } j)\times(\text{quanity of variable } j).$$

At optimum,

$$\text{Money}=\sum_{i=1}^{m}(\text{quanity of resource } i)\times(\text{ unit worth of resource } i).$$

Hence,

$y_i \equiv$ Worth per unit of resource $i \equiv$ Dual price $\equiv$ Shadow price

$\equiv$ Simplex multiplier

From Property II

$$(\text{Objective coefficient of } x_j) = \sum_{i=1}^{m} a_{ij} y_i - c_j$$

At any simplex iteration of the primal or the dual, Objective coefficient of variable *j* in one problem= Left-hand side minus right-hand side of constraint *j* in the other problem

Interpretation

- As objective function denotes profit per unit,

$$\sum_{i=1}^{m} a_{ij} y_i \Rightarrow \text{money/unit}$$

$$c_j \rightarrow \text{profit per unit}$$

$$a_{ij} \rightarrow \text{amount of resource } i \text{ used by variable } j$$

Hence,

$$y_i \rightarrow \text{Imputed cost per unit of resource } i$$

An imputed cost, also known as an implicit cost, implied cost, or opportunity cost, represents the value of a resource that is used by a business but not directly paid for.

Interpretation

$$\sum_{i=1}^{m} a_{ij} y_i \Rightarrow \text{Imputed cost of all the resources in making product } j$$

$$\text{Pay attention to } \sum_{i=1}^{m} a_{ij} y_i - c_j . \text{If it is negative, we can increase } x_j .$$

$$\text{There is logic in enhancing the production, as long as}$$

$$\left(\text{Imputed cost of used resources for product j}\right) < \left(\text{profit per unit of product j}\right).$$

Interpretation

$$z_j = \sum_{i=1}^{m} a_{ij} y_i$$

$z_j - c_j$ is objective coefficient of x_j in the simplex table.

It is called reduced cost of variable j.

Maximize $z = 50x_1 + 40x_2 + 30x_3 + 25x_4 + 20x_5 + 10x_6$

subject to

$$5x_1 + 2x_2 + x_3 + x_4 + x_5 + 2x_6 \le 10,$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \ge 0.$$

Dual form:

Minimize $w = 10y$

Subject to

$5y \ge 50,$

$2y \ge 40,$

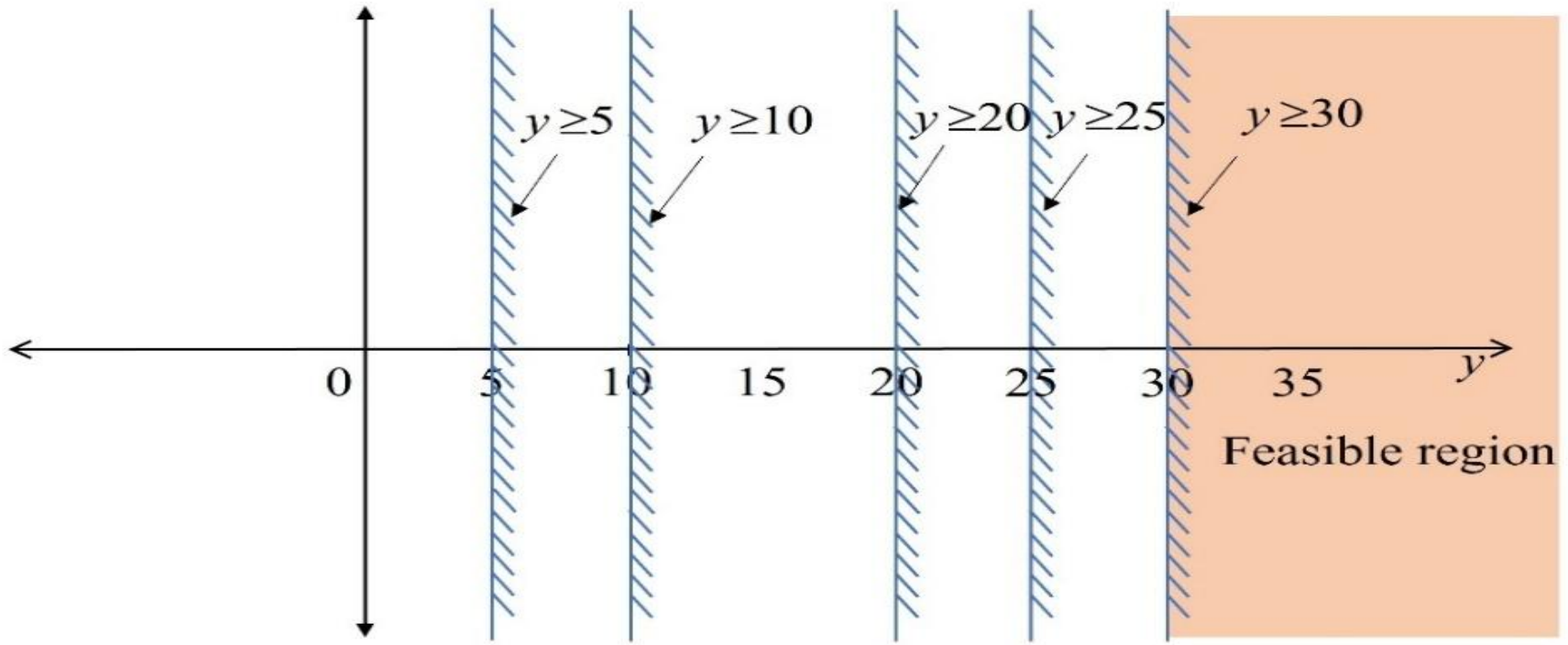
$y \ge 30,$

$y \ge 25,$

$y \ge 20,$

$2y \ge 10,$

$y \ge 0.$



The feasible region for y is bounded by $y \geq 30$, as it is the tightest bound. Therefore, the feasible region includes all values of $y \geq 30$.

The objective function is $w = 10y$. To minimize w, we need to take the smallest possible value of y within the feasible region ($y = 30$). Hence, minimum $w = 10 \times 30 = 300$.

At the optimum, the objective value of dual and corresponding primal problem is equal. Hence, maximum value of $z = 300$.

Dual constraint	Primal variable
5y = 5 × 30 = 150 > 50 (not tight)	$\Rightarrow x_1 = 0$ (corresponds to x_1 in primal problem)
2y = 2 × 30 = 60 > 40 (not tight)	$\Rightarrow x_2 = 0$ (corresponds to x_2 in primal problem)
y = 30 = 30 (tight)	$\Rightarrow x_3 \neq 0$ (corresponds to x_3 in primal problem)
y = 30 >25 (not tight)	$\Rightarrow x_4 = 0$ (corresponds to x_4 in primal problem)
y = 30 > 20 (not tight)	$\Rightarrow x_5 = 0$ (corresponds to x_5 in primal problem)
2y = 2 × 30 = 60 > 10 (not tight)	$\Rightarrow x_6 = 0$ (corresponds to x_6 in primal problem)

Since only the constraint $y \geq 30$ is tight, the corresponding primal variable x_3 is positive.

Now, from the primal constraint:

$5x_1 + 2x_2 + x_3 + x_4 + x_5 + 2x_6 = 10$

$\Rightarrow 5 \times 0 + 2 \times 0 + x_3 + 0 + 0 + 2 \times 0 = 10$

$\Rightarrow x_3 = 10.$

Hence, optimal primal solution is $(x_1, x_2, x_3, x_4, x_5, x_6) = (0, 0, 10, 0, 0, 0)$.

Unit worth of resource

As y=30, unit worth of resource of constraint

$$5x_1 + 2x_2 + x_3 + x_4 + x_5 + 2x_6 \leq 10$$

is 30. That means if rhs changes to 11, the objective function will Increase by 30.

$$z = 50x_1 + 40x_2 + 30x_3 + 25x_4 + 20x_5 + 10x_6$$

Solution will be x_3 =11 and z will be 330.